ECEN 470 Exam #1 Review

1. Know the form of a feedback system and the names associated with various components in the block diagram.
2. Given that the input is $\alpha e^{s_1 t}$ to a LTI system with impulse response, $H(s)$ what is the output $y(t)$? (See Section 3.1.2, page 96-97, equation (3.17))
3. Understand the relationship between rise time, settling time, overshoot and the location of poles 2nd order poles in the complex plane.
4. Know how the response of a 2nd order system is changed by adding a zero or pole at various positions on the real axis.
5. Know the equation for system error.
6. Be able to find the response (e.g., rise time) of a DC motor to a step input.
7. Given a LTI system that has unity feedback, be able to find the range of proportional feedback for which the system is stable.
8. Be able to apply the initial and final value theorem.
9. Know the definition and characteristics of a non-minimal phase system. (section 3.5, page 144-145, Example 3.30, Problem: 3.46 (c))
10. Be able to take the Laplace transform of a function.
11. Understand the equation for the Law of Motors (Sec. 2.3, Eq. 2.53) and the Law of Generators (Sec. 2.3, Eq. 2.56).
12. Be able to derive the equations of motion for a rotational mass, spring, damper system. (Sec. 2.1.2)
13. Be able to calculate math equations expressed in the form of a block diagram.
14. Be able to apply partial fraction decomposition.
15. Be able to find the inverse Laplace transform using look up tables.
16. Be able to find the equation of translational motion for a mass, damper, spring system.
17. Know the form of the transfer function for a DC motor where angular velocity is the output and applied armature voltage is the input.
18. What does a row of zeros in the Routh array tells you about the location of the roots?
19. Know the response caused by having a pole at the origin. (Hint: consider doing a partial fraction decomposition and then taking the inverse Laplace.)
20. Know the properties of integrating an impulse multiplied by a function, $\int_{-\infty}^{\infty} f(t)\delta(t-t_o)dt$.
21. Know why it is NOT a good idea to cancel a pole in the right have plane by placing a zero on top of it.
22. Understand the relationship between the convolution integral and the impulse response. And, representing an arbitrary input as the sum of weighted impulses. (see Section 3.1.1 and Fig. 3.2)
23. Understand the relationship between the roots of the characteristic equation and stability.
24. Be able to determine if the roots of a characteristic equation are in the right-half plane.
25. Be able to construct and interpret a Routh array.
26. Know Newton's law for rotational motion (Sec. 2.1.2, Eq. (2.10)).
27. Know the type of response obtained because of pole location(s) in the complex plane.
28. Know and be able to apply the final value theorem.

29. Know the equation for back emf in a DC motor (Sec. 2.3.2, Eq. 2.61).
30. Know what equation for the torque produced by a DC motor (Sec. 2.3.2, Eq. 2.60).